

EFU MASTER INDEX – ENGLISH CONSOLIDATED VERSION (v3.1)

0. Genealogy – Where EFU Comes From

EFU emerges from a sequence of research essays that move from ethical and ecological reflection to a quantitative ontology of life and, finally, to a human-scaled measurement system. Early work on human responsibility within ecosystems and learned helplessness introduces the idea of structural metabolic injustice and collective paralysis. The Nookratic Alliance texts define a space for collective intelligence and AI-assisted governance, where human dignity and planetary limits must co-evolve.[[simoristvan.wordpress](https://simoristvan.wordpress.com)]

This culminates in a quantitative ontology of life, which describes matter, information, consciousness and networks as measurable layers of a single flux system. From this basis, the EFU – Human Flux Unit paper defines 1 EFU-H as the reference metabolic throughput of an average human, creating a common scale for comparing any industrial, digital or institutional process to human life-support. The EFU Master Index is the institutionalisation of this trajectory into a coherent family of standards, audits and governance tools.

I. METABOLIC FOUNDATIONS (104.x – Core EFU Dimensions)

This block defines the core EFU dimensions that quantify key biogeochemical and energetic fluxes on a human-equivalent scale.

- **104.5 EFU-C (Carbon)**
Baseline: 1 tonne CO₂-equivalent.
Alignment: ISO 14064, GHG Protocol, IPCC.
Role: Additional human-equivalent CO₂e layer on top of existing carbon accounting.
Status: Official v1.0.
- **104.6 EFU-W (Water)**
Baseline: 18.25 m³ per person per year (50 l/day).
Alignment: ISO 14046, WHO guidelines.
Role: Regional water-scarcity and RCC (regional carrying capacity) audits in human terms.
Status: Official v1.0.
- **104.7 EFU-N (Nitrogen)**
Baseline: Reactive nitrogen flux baseline.
Alignment: EU Nitrates Directive 91/676/EEC.
Role: Flux-based reactive nitrogen accounting.
Status: Official v1.0.
- **104.8 EFU-H (Energy)**
Baseline: 12.88 MWh per person per year (~1.47 kW continuous), derived from human metabolism.
Alignment: ISO 50001.

Role: Anthropocentric energy baseline for all energy systems.
Status: Official v1.0.

- **104.9 EFU-S (Entropy)**

Baseline: 11 500 J/K per year (human metabolic entropy).
Alignment: Second law of thermodynamics.
Role: System-level thermodynamic dissipation metric.
Status: Official v1.0.

- **104.10 EFU-P (Phosphorus)**

Baseline: Phosphorus flux baseline.
Alignment: EU Fertilising Products Regulation 2019/1009.
Role: Circular phosphorus flux assessment.
Status: Official v1.0.

- **104.11 EFU-Pr (Protium / H₂)**

Baseline: Hydrogen flux baseline.
Alignment: ISO 14001, hydrogen strategies.
Role: Hydrogen-specific flux audits.
Status: Official v1.0.

- **104.12 EFU-Kr (Cryptocurrency)**

Baseline: Bitcoin ~190 TWh/year as a reference.
Alignment: Thermodynamics of Proof-of-Work.
Role: Digital-physical energy bridge, with MROI thresholds (unconstitutional $<10^{-7}$).
Status: Official v1.0 (~170 pages).

- **104.13 EFU-O₂ (Oxygen)**

Baseline: 230 kg O₂ per person per year (~630 g/day).
Alignment: ISO 14001, 14046, 14064.
Role: Atmospheric O₂ consumption; aquatic DO deficits expressed in human-equivalent terms.
Status: Official v1.0 (2026-01-12).

Note: In future versions (v4.0+), the order may be rearranged to follow biogeochemical logic (C, W, N, O₂, P, S, H/Pr, Energy, Crypto).

II. SECTORAL AUDIT PROTOCOLS (104.x – Applied Licensing)

These modules embed EFU metrics into concrete sectoral audits and licences.

2.1 Primary production and industry

- 104.14 Forestry – EFU-C sequestration, EFU-O₂ production, EFU-W transpiration; aligned with ISO 14001.
- 104.15 Food industry – EFU-H, EFU-W, EFU-C across supply chains; ISO 22000, GlobalG.A.P.
- 104.19 Fisheries & aquaculture – EFU-W, EFU-N, EFU-O₂ (DO conflicts); MSC/ASC.
- 104.20 Mining – EFU-H, EFU-W, EFU-M (ore EFU extraction); ISO 14001, ICMM.
- 104.23 Agriculture – EFU-N, EFU-P, EFU-W, EFU-H; ISO 14001, 50001, EU CAP.

- 104.27 Animal husbandry – EFU-H, EFU-W, EFU-C, EFU-N (animal metabolism).
- 104.28 Chemical industry – EFU-H, EFU-C, EFU-S (process energy / entropy).
- 104.32 Metallurgy – EFU-H, EFU-C, EFU-O₂ (furnace oxygen demand).
- 104.33 Construction – EFU-M, EFU-H, EFU-C (embodied energy and materials).

2.2 Logistics and infrastructure

- 104.24 Fossil fuels – EFU-C, EFU-H, EFU-O₂ (combustion).
- 104.25 Shipping – EFU-C, EFU-H, EFU-O₂ (bunker fuel); MARPOL, IMO GHG.
- 104.26a Land transport – EFU-H, EFU-C, EFU-O₂; ISO 14083, EU EURO norms.
- 104.29 Aviation – EFU-H, EFU-C, EFU-O₂; ISO 14083, CORSIA.
- 104.30 Waste management – EFU-M, EFU-H, EFU-N; EU Waste Framework Directive.
- 104.31 Energy utilities – EFU-H, EFU-C, EFU-S (grid balancing).
- 104.34 Wastewater – EFU-W, EFU-N, EFU-O₂ (BOD/COD, DO deficits).

2.3 Digital and socio-technical systems

- 104.4 Human systems – EFU-H (cognitive load), EFU-W (baseline); ISO 30414.
- 104.12 Cryptocurrencies – EFU-H, EFU-S, EFU-W, EFU-M, MROI; PoW thermodynamics.
- 104.37 Digital metabolism (AI) – EFU-H (compute), EFU-W (cooling), EFU-S (GPU entropy); EU AI Act, ISO/IEC 42001.
- 104.42 Social media – EFU-H (attention-energy, cognitive burden); EU DSA.
- 104.43 Mobile ecosystem – EFU-M (e-waste), EFU-H (device life-cycle); ISO 14001, EU WEEE.

III. GOVERNANCE AND DATA PROTOCOLS (200-Series)

These documents define the institutional mapping, software stack and data interfaces for EFU-based audits.

- **201 EFU Glossary** – Universal terminology, SI units, fixed bio-physical constants (e.g. EFU-H = 12.88 MWh/year).
- **202 Institutional Mapping Protocol** – 5-step “Physical GAAP” audit (M_{inst}, M_{allocated}, SS, MROI); aligned with EU CSRD.
- **203 Case Studies** – Cities (e.g. Paris, Abu Dhabi) and industrial pilots illustrating EFU audits.
- **204 Calculator Logic Overview** – Mathematical algorithms and bio-physical conversions.
- **205.1 Software Specification** – XBRL-ready architecture, repository structure.
- **205.2 Data Model and Logical Relations** – ER diagrams, relational schema (projects, resources, audits).
- **205.3 JSON Schema Specification** – Schema and validation rules, REST interface.
- **205.4 API Documentation** – OpenAPI 3.0 endpoints for machine-to-machine communication (draft).

- **207 Audit Layer Note** – Third-party audit methodology, consistent with ISO 19011.

IV. STRATEGIC AND PHILOSOPHICAL PILLAR (100-Series – Policy Layer)

This block holds the normative, conceptual and strategic backbone of EFU.

- **101 EFU Manifesto** – Six ethical pillars: human compass, creativity, ethics, future awareness, community well-being, sovereignty (HU/EN).
- **102 EFU Study** – Scientific foundations: bio-physical basis, “metabolic predator”, “sovereignty gap”.
- **103 Executive Summaries** – One-pager and extended brief for decision-makers (national and institutional).
- **104 Licensing System (Folder)** – Legal architecture for all EFU modules.
 - 104.0 Universal Licence (EFU-UNIV) – Umbrella agreement for all EFU modules.
 - 104.1 Software Licence (Apache 2.0) – Open integration of the EFU calculator.
 - 104.2 Content Licence (CC BY 4.0) – Intellectual property and attribution (Simor István).
 - 104.3 Use Guidelines – Brand protection, “EFU-Sovereign” label, anti-greenwashing rules.
- **105 Fundamental Axiom** – “Measurement is a precondition for sovereignty.”
- **106 Physical Sovereignty and the Future of AI** – Digital growth constrained by bio-physical limits.
- **107 Global Positioning and Consequences** – Integration with ESG, CSRD, EU AI Act and similar frameworks.
- **108 Sovereignty Pilot Report v1.0** – Summary of initial field tests.
- **109 Digital Infrastructure Metabolism** – Cloud/data-centre physical transformation.
- **110 Interstitium – EFU Bio-Empirical Foundation** – Human body’s largest flux field as proof-of-concept (HU/EN).
- **111 Extension of the EFU Concept** – Professional development of the “Quantitative Ontology of Life”.
- **112 Conceptual Foundations and Definitions** – Quantitative Ontology terminology.
- **113 Two Main Research Questions (K1, K2)** – Information stability vs. matter-flux tension (“quantitative shock”).
- **114 EFU Strategic Roadmap** – DOI registration, ISO/UN integration path.
- **115 EFU Carbon Standard v1.0** – Additional audit layer on ISO 14064 / GHG Protocol.
- **116 Legal and Research Status Statement** – EFU as research hypothesis and experimental framework.
- **117 EFU-Energy Summary** – Translating energy consumption into human dignity units.
- **118 Human Flux Unit (EFU) – Landing Page** – Global entry point; “Shares vs. Flux” paradigm shift.
- **119.1 Goals and Positioning** – Framework objectives and scientific stance.

- **120 ISO Compliance and Integration Strategy** – Route towards formal standardisation.

V. SPECIAL MODULES AND EXTENSIONS

These modules explore frontier applications and meta-systemic dimensions of EFU.

- **EFU-Protium** – Human metabolic baseline (EFU-H); 1 EFU-H = 12.88 MWh/year.
 - **EFU-Entropy** – Thermodynamic dissipation standard (EFU-S); 1 EFU-S = 11 500 J/K/year.
 - **EFU-Crypto** – Proof-of-Work audit (EFU-Kr); Bitcoin MROI $\approx 2.99 \times 10^{-8}$, 8 EFU axioms.
 - **EFU-Carbon** – Flux-based CO₂ accounting (EFU-C), integrating Scope 1–2–3.
 - **EFU-Oxygen** – Atmospheric and aquatic O₂ flux (EFU-O₂); DO deficits as human-equivalents.
 - **Interstitium** – Bio-empirical foundation: 10 L human interstitial space as largest internal flux field, including trauma-flux hypothesis.
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VI. KEY FORMULAS AND CLASSIFICATION THRESHOLDS

These metrics translate complex bio-physical and socio-economic states into clear governance signals.

1. Sovereignty Gap (SS)

- Formula:

$$SS = M_{inst} - M_{allocated}$$
 where

$$M_{allocated} = N_{staff} \times RCC$$
- Classification:

$$SS \leq 0 \rightarrow \text{Sovereign}$$

$$0 < SS \leq \text{threshold} \rightarrow \text{Balanced}$$

$$SS > \text{threshold} \rightarrow \text{Deficit / Critical}$$
- Purpose:
 To trigger interventions when institutional consumption exceeds the sustainable regional capacity allocated to that institution.

2. Total EFU Cost (TEFU)

- Formula:

$$TEFU = w_H \times EFU_H + w_W \times EFU_W + w_M \times EFU_M + w_S \times EFU_S + w_C \times EFU_C$$
- Default weights:

$$w_H = 1.0$$

$$w_W = 1.0$$

$$w_M = 0.5$$

$$w_S = 0.8$$

$$w_C = 0.3$$

- Purpose:
To aggregate all relevant EFU dimensions into a single metabolic cost indicator for a project, institution or sector.

3. Metabolic Return on Investment (MROI)

- Formula:
 $MROI = THI / TEFU$
- THI definition:
 $THI = 1.0 \times \text{Jobs}$
 - $0.1 \times \text{GDP}$
 - $0.5 \times (\text{Users} / 1\,000)$
 - $2.0 \times \text{Innovation}$
- Thresholds:
 $MROI > 1E-3 \rightarrow \text{green (excellent)}$
 $MROI > 1E-4 \rightarrow \text{yellow (acceptable)}$
 $MROI > 1E-5 \rightarrow \text{orange (problematic)}$
 $MROI < 1E-7 \rightarrow \text{red (unconstitutional)}$
- Purpose:
To measure societal usefulness generated per unit of metabolic cost.

4. Governability (G)

- Formula:
 $G = (\Delta R / \Delta I) \times T \times C$
- Parameter meanings:
 $\Delta R / \Delta I$ = response-to-input ratio
 T = transparency
 C = community participation
- Thresholds:
 $G > 1.0 \rightarrow \text{high governability}$
 $G < 0.5 \rightarrow \text{governance crisis}$
- Purpose:
To assess a system's ability to convert inputs (signals, demands, shocks) into meaningful, timely and transparent responses.

VII. DOCUMENT STATUS LEGEND

- Official v1.0 – Published, ready for peer review and DOI.
 - Draft / Concept – Under development, not final.
 - Missing – Identified gap, needs development.
 - Active research – Ongoing empirical validation.
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